

Module 2 Assignment

Student Record Manager

Java Program

Features: Add Student • Validate Email using Regex • Save Data to File • Read Student Data • Handle Invalid Input using Exceptions

Complete Program

```
import java.io.*;
import java.util.*;
import java.util.regex.*;

class Student {
    int id;
    String name;
    String email;

    Student(int id, String name, String email) {
        this.id = id;
        this.name = name;
        this.email = email;
    }

    @Override
    public String toString() {
        return id + "," + name + "," + email;
    }
}

public class StudentRecordManager {

    static final String FILE_NAME = "students.txt";
    static Scanner sc = new Scanner(System.in);

    // Validate Email using Regex
    static boolean isValidEmail(String email) {
        String regex = "^[A-Za-z0-9+_.-]+@[A-Za-z0-9.-]+$";
        return Pattern.matches(regex, email);
    }

    // Add Student
    static void addStudent() {
        try {
            System.out.print("Enter Student ID: ");
            int id = Integer.parseInt(sc.nextLine());

            System.out.print("Enter Student Name: ");
            String name = sc.nextLine();

            System.out.print("Enter Email: ");
            String email = sc.nextLine();

            if (!isValidEmail(email)) {
                throw new IllegalArgumentException("Invalid email format!");
            }

            Student student = new Student(id, name, email);

            // Save data to file
            FileWriter fw = new FileWriter(FILE_NAME, true);
            BufferedWriter bw = new BufferedWriter(fw);

            bw.write(student.toString());
            bw.newLine();

            bw.close();

            System.out.println("Student added successfully!");

        } catch (NumberFormatException e) {
            System.out.println("Invalid input! Student ID must be a number.");
        } catch (IllegalArgumentException e) {
            System.out.println(e.getMessage());
        } catch (IOException e) {
        }
    }
}
```

```

        System.out.println("Error while saving student data.");
    }
}

// Read Student Data
static void readStudents() {
    try {
        File file = new File(FILE_NAME);

        if (!file.exists()) {
            System.out.println("No student records found.");
            return;
        }

        BufferedReader br = new BufferedReader(new FileReader(file));

        String line;
        System.out.println("\n--- Student Records ---");

        while ((line = br.readLine()) != null) {
            String[] data = line.split(",");

            System.out.println("ID      : " + data[0]);
            System.out.println("Name   : " + data[1]);
            System.out.println("Email  : " + data[2]);
            System.out.println("-----");
        }

        br.close();

    } catch (IOException e) {
        System.out.println("Error while reading student data.");
    }
}

// Main Method
public static void main(String[] args) {

    while (true) {
        System.out.println("\n===== Student Record Manager =====");
        System.out.println("1. Add Student");
        System.out.println("2. Read Student Data");
        System.out.println("3. Exit");

        try {
            System.out.print("Enter your choice: ");
            int choice = Integer.parseInt(sc.nextLine());

            switch (choice) {
                case 1:
                    addStudent();
                    break;

                case 2:
                    readStudents();
                    break;

                case 3:
                    System.out.println("Program exited.");
                    sc.close();
                    return;

                default:
                    System.out.println("Invalid choice! Please select 1, 2 or 3.");
            }

        } catch (NumberFormatException e) {
            System.out.println("Invalid input! Please enter a number.");
        }
    }
}
}

```

GitHub Repository Structure

```

Student-Record-Manager/
■■■ StudentRecordManager.java
■■■ students.txt
■■■ README.md

```